

Network Security Assessment

Target: sample_scan.xml
Scanned: 2026-08-04T11:36:24
Nmap invocation: `nmap -sT -sV -F -oX scan.xml scanme.nmap.org`
Hosts up: 1
Rule-engine findings: high: 1 · medium: 2
Analyst: llama-3.3-70b-versatile (Groq)
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This assessment covers an authorised scan only. Version-based findings are inferred from service banners and require verification against vendor advisories before being treated as confirmed vulnerabilities.

Executive Summary

The scan of scanme.nmap.org reveals a host with two open ports, indicating a potential exposure of services to the internet. The most urgent issue is the outdated OpenSSH software, which poses a significant risk due to known user-enumeration and key-handling issues. This host appears to be intentionally public, given the presence of an HTTP server, but the lack of HTTPS and outdated software versions suggest a need for attention to security configuration. Overall, the exposure of this host warrants prompt remediation to prevent potential exploitation.

Risk Assessment by Host

The host scanme.nmap.org presents a mixed posture, with both expected and concerning findings. On one hand, the presence of an HTTP server suggests a deliberate exposure to the public. On the other hand, the outdated software versions and lack of HTTPS raise concerns about the security of the host.

Severity	Finding	Port	Why it matters
HIGH	Outdated OpenSSH	22	Enables potential user-enumeration and key-handling issues
MEDIUM	Outdated Apache httpd	80	Misses a long run of security fixes, potentially exposing the host to known vulnerabilities
MEDIUM	HTTP served without HTTPS	80	Sessions and credentials travel in the clear, risking interception and eavesdropping

Suspicious Indicators

The host scanme.nmap.org does not exhibit overtly suspicious indicators, such as unexpected open ports or unusual service banners. However, the lack of HTTPS on the HTTP server and the outdated software versions may suggest a lack of attention to security configuration or maintenance.

Likely Vulnerabilities

Based on the detected software and versions, the following vulnerability classes are implied:

- User-enumeration and key-handling issues in OpenSSH (CONFIRMED)
- Various security fixes missed in Apache httpd 2.4.x below 2.4.40 (REQUIRES VERIFICATION, as the scan only provides version information and not direct evidence of exploitability)
- Clear-text transmission of sessions and credentials due to the lack of HTTPS (CONFIRMED)

Prioritised Remediation

1. **Update OpenSSH to a version 7.4 or later** on scanme.nmap.org:22 (medium effort) to address known user-enumeration and key-handling issues. 2. **Enable HTTPS on the HTTP server** on scanme.nmap.org:80 (low effort) to protect sessions and credentials from interception and eavesdropping. 3. **Update Apache httpd to a version 2.4.40 or later** on scanme.nmap.org:80

(medium effort) to apply a long run of security fixes and mitigate potential vulnerabilities.